Trøndelagkonkurranse 2026 - Level I

Second Round - November 1, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Find the second decimal place after the comma in the decimal expansion of

$$\frac{1}{1+\sqrt{101}+\sqrt{103}}$$

Note: The second decimal place after the comma in π is 4.

Problem 2

Frida writes each of the numbers $1, 2, 3, \ldots, 2026$ on a blackboard once. At each step, she can erase two numbers on the blackboard, say x and y, and replace them with the single number $|x-y|$. After several steps, there is only one number on the blackboard. Is it possible that she obtains 0 at the end?

Problem 1

Since $100 < 101 < 121$ and $100 < 103 < 121$, it follows that

$$10 < \sqrt{101} < 11 \quad \text{and} \quad 10 < \sqrt{103} < 11.$$

Adding up the inequalities and adding 1 we obtain

$$20 < \sqrt{101} + \sqrt{103} < 22$$

$$21 < 1 + \sqrt{101} + \sqrt{103} < 23.$$

Furthermore, we can widen the bounds as

$$20 < 1 + \sqrt{101} + \sqrt{103} < 25.$$

Inverting we obtain

$$\frac{1}{20} > \frac{1}{1 + \sqrt{101} + \sqrt{103}} > \frac{1}{25},$$

which is equivalent to

$$0.05 > \frac{1}{1 + \sqrt{101} + \sqrt{103}} > 0.04,$$

thus the decimal expansion of the fraction is 4.

Problem 2

It is NOT possible. See problem 2 of Level II.